PNAS CORE CONCEPTS

Can neuromorphic computing help reduce AI's high energy cost?

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Some researchers believe that a computing architecture inspired by the mechanisms of the brain could offer a path to innovation that helps tackle rampant AI energy consumption. Image credit: Shutterstock/MDV Edwards.

Artificial intelligence systems, and large language models (LLMs) in particular, have become almost ubiquitous. The implications are many. Among them: serious concerns about massive amounts of energy usage. The initial training of GPT-3 required as much energy as powering 120 houses for a year. That's before adding the energy required for the chatbot to respond to users' prompts (1–3). GPT-4 required an estimated 50 times as much energy as its predecessor to train, and although OpenAI has not divulged official figures, GPT-5 likely required much more.

To date, most efforts to improve the energy consumption of AI systems have focused on developing more efficient algorithms, increasing the use of energy from sustainable sources, or even building smaller language models. To build DeepSeek, which debuted last January, researchers in China built a model that only activates a fraction of the total model for each query. Other efforts focus on packing more transistors onto smaller chips, which results in a shorter distance for the data to travel and enables more parallel computing.

But many researchers are developing a different kind of computing architecture, one inspired by the efficient mechanisms of the brain. Dubbed neuromorphic computing, it was first proposed decades ago but has recently seen a resurgence of

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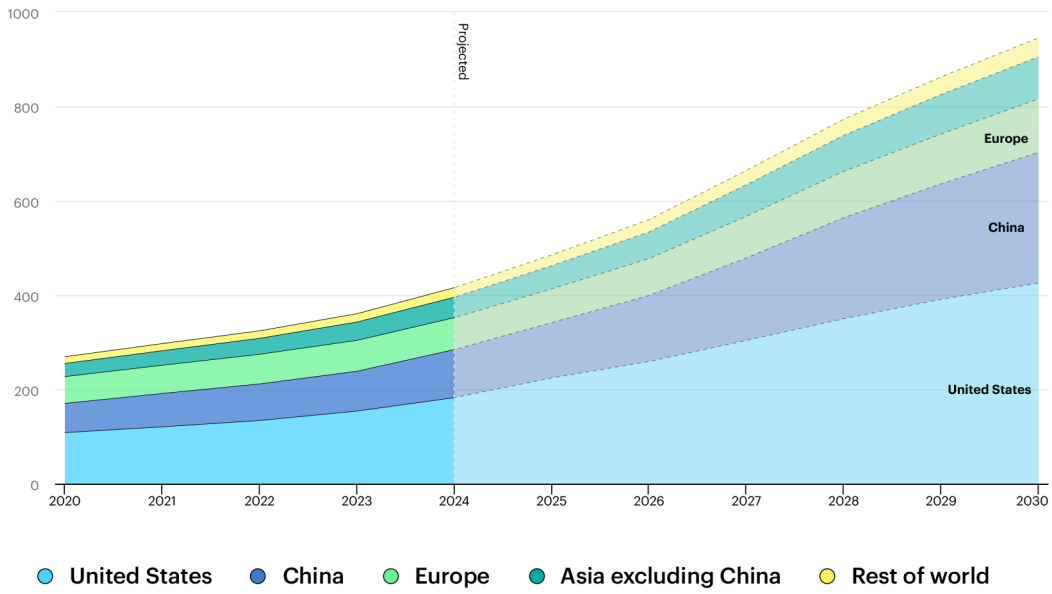

From 2020 to 2030, data center electricity consumption will continue to vary greatly by region, according to projections from the International Energy Association (IEA). Image credit: IEA, licensed under CC BY 4.0.

interest. Some believe neuromorphic approaches could offer a path to innovation that helps tackle the formidable problem of rampant AI energy consumption.

Brainy Inspiration

The appeal is undeniable: For all its cognitive tasks, the human nervous system runs on roughly 20 watts, the same power demand as a couple of standard LED bulbs (4). Since the late 1980s, researchers have proposed designing hardware and algorithms that closely mimic what we know about the configuration of neurons and synapses. This concept, neuromorphic computing, relies on a network of artificial neurons to crunch data as events unfold, rather than continuously, in the same location as memory—and with parallel computing architecture. Today's high-powered AI chips also use massive parallelism by running many processing units at once, but this brute force method of breaking up large tasks into smaller ones requires enormous stores of energy.

Many researchers now view neuromorphic computing as an appealing alternative to packing more processors into ever-tinier spaces. This approach could conserve energy in computation or, when used with existing systems, offset the energy use of existing frameworks. Thus far, the energy savings and other benefits aren't substantial enough to attract large companies, especially those that have invested heavily in other AI architectures. Still, the approach could lead to new engineering insights and unearth niche applications that conventional computing does not replicate well, such as modeling the connections among the tens of billions of neurons in the brain.

"Neuromorphic has gained traction recently because of the gains in artificial intelligence and the need for lots of energy and data centers," says Grace Hwang, a computational neuroscientist at the NIH in Bethesda, Maryland. "Engineers are looking to neuroscience to say, 'hey, the brain is very efficient. What are we missing here?'"

That inspiration is making its way to tech markets, at least for purposes of demonstration and experimentation. The research group of engineer Tobi Delbruck, at ETH Zurich and the University of Zurich in Switzerland, for example, has produced neuromorphic-based artificial retinas and cochleas, to be used in computing systems for image or sound recognition, that emulate the mechanisms of their biological counterparts. Other projects focus on computing: Several major companies have recently produced neuromorphic computer chips, some of which reportedly perform AI tasks, such as object recognition, using less energy than conventional processing units. In May 2024, a German company called SpiNNcloud partnered with Sandia National Laboratories in Albuquerque, New Mexico, to investigate how neuromorphic chips might bolster national defense systems (5). In September 2021, Intel produced Loihi 2, a neuromorphic chip that simulates over 1 billion neurons (6). And in October 2023, IBM reported that its latest version of a neuromorphic chip, NorthPole, outperformed conventional computing architectures on a variety of tasks, and at a fraction of the energy cost (7). NorthPole is an "AI Accelerator' that's "designed with energy efficiency in mind," says Dharmendra Modha, IBM's chief scientist for brain-inspired computing.

Computing Like the Brain

The idea for neuromorphic computing originated with Carver Mead, an engineer and computer scientist at the California Institute of Technology, in Pasadena. In the 1980s, Mead cultivated an intense interest in the physics of computation and popularized the term "Moore's Law"—the notion, first introduced by Intel co-founder Gordon Moore in 1965, that due to innovation, the number of transistors on an integrated circuit will double every two years (8).

Driven by an increasing fascination with the nervous system of animals, Mead began to model the nervous system in silicon. With his collaborators and students, he used analog electrical circuits to create virtual hearing and vision systems;

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in 1992, they produced a neuromorphic, bio-inspired sensing device called a silicon retina, or "event camera," designed to mimic the human eye (9). Since these early steps, the field has evolved and grown. Researchers have continued to develop and improve sensing devices—such as those in Delbruck's lab—and work on uses in robots.

In the last couple of decades, the word "neuromorphic" has taken on broader meanings in computing, says computer scientist Dhireesha Kudithipudi at the University of Texas at San Antonio. "Taking principles of the neural processes or abstracting them into various degrees, whether in hardware or software, is still considered neuromorphic," she explains.

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>
> **—Catherine Schuman**

But other researchers, including those more focused on saving energy, are looking at ways to mimic specific processes and mechanisms that make the brain more efficient and integrate them into broader computing systems. That's true for Modha's group at IBM. "We are driven not so much by neuroscience, but more by the intrinsic mathematical potential of the architecture," he says.

All Together Now

Today's AI systems, even though they're driven by neural networks originally inspired by the structure of the brain, bear little resemblance to their biological counterparts.

They train on enormous stores of data before they can produce accurate predictive results from queries, and most machine learning models don't retain previously learned knowledge as they move on to new tasks. And they're built on systems that run continuously. They keep memory and processing components separate. A hard drive stores data. The RAM, or random-access memory, contains data that are needed by the processing unit as it runs software. This is not how the brain functions.

Perhaps most crucially, humans are lifelong learners, says Kudithipudi, noting that "we are not able to see this in artificial systems." In a 2022 perspective article, she and her colleagues described biological mechanisms associated with lifelong learning—and offered ways that those mechanisms might be incorporated into neuromorphic approaches (10). People learn continually throughout their lifetimes without new hardware, she notes. Past experiences inform how we absorb new information. "A child sees an example of something a couple of times, and then they just learn, 'this is a cat; this is a dog'," says electrical engineer Adnan Mehonic at University College London. "They don't need to see a dog 1 million times to know it's a dog."

Continuous learning is possible, says Kudithipudi, in part because memory and processing happen together in the brain. That's become a hallmark of neuromorphic computing architectures: Neuromorphic chips have memory built in. Many use a device called a memristor—a portmanteau of "memory" and "transistor"—that acts as both a processing and a memory component. Combining those two functions means the system doesn't have to use energy to move data from a hard drive to the RAM. Plus, neuromorphic chips are event-driven, which means they use energy only while they're completing a task.

Eliminating the steps required to transfer data between memory and processing components saves energy and time. In a 2023 paper, Modha and his team at IBM reported that the NorthPole neuromorphic chip successfully classified images from a dataset—a task often used to benchmark the performance of AI systems. The chip did so using a tiny fraction of the energy required by a conventional system, and it was five times faster (7). Modha believes that building chips differently, rather than only finding ways to shrink circuit dimensions and pack more processors onto integrated circuits, can lead to greater gains in energy efficiency. "Architecture trumps Moore's Law," he says.

Just in Time

In addition to co-locating memory and processing, researchers are investigating ways to mimic the timing mechanisms of the brain. Since the 1950s, neuroscientists have modeled communication between neurons as a spiking neural network (SNN). These "spikes" are sudden surges in the voltage of an individual neuron, typically lasting 2–5 milliseconds and triggered by changes on the cell membranes as neurons exchange signals with each other (11). The spikes travel along the neuron and can strengthen the synapses.

Information may be encoded in the firing patterns of these spikes, whether in the individual neurons or in patterns that show up in networks of neurons. This sparse, rather than continuous, use of electricity is a key factor in their energy efficiency (12). If the output for some computation is zero, then the neuron doesn't fire. Scientists still don't understand fine-grained details, such as how a specific thought, memory, or reasoning process is encoded in these timed electrical spikes. Nonetheless, many neuromorphic hardware and software research groups are investigating ways to mimic these spikes and neurons.

In theory, spiking neural networks should be able to complete most of the same tasks as other architectures, says computer scientist Catherine Schuman at the University of Tennessee, Knoxville. And they should do so using less energy. "You ought to be able to implement the same computation with fewer elements," Schuman notes. But thus far, there has been limited success in training SNNs to outperform conventional neural networks (12). Some studies suggest that it's a matter of finding the best rules to train SNNs; others point to ways to improve the timing of the spikes. Schuman's group is focused on developing SNNs that are energy-efficient, scalable, and can leverage the benefits of other techniques like reinforcement deep learning. She's particularly interested in small networks. "We can get very small networks doing relatively complex tasks," she says. "And in the way that it operates, it's more energy-efficient."

Applications and Challenges

Energy gains fuel much of today's scientific interest in neuromorphic software and hardware. But for many researchers,

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the most exciting application may be the ability to model the brain at unprecedented scales in hopes of better understanding how human brains process and store information.

The European Union's Human Brain Project, which began in 2011 and ended in 2023, produced an open-source, neuromorphic platform called BrainScaleS-2 that uses an SNN to emulate a brain processing information in real time (13, 14). Its artificial neurons are connected by plastic synapses, and it uses physical quantities like current and voltage to represent the dynamic interactions between neurons (14). Brain Project researchers released a small-scale hardware device, also using BrainScaleS-2, that doesn't require a host machine—potentially opening the door to wearables or sensors, for example. High-speed robotics applications, neuroprosthetic devices and new wearable devices are also possibilities (15–17). And Hwang points to Intel's neuromorphic Loihi 2 chip, which simulates about 1.15 billion neurons. "This allows theorists to ask questions and make observations that are only available at large scales," she says.

Neuromorphic architectures also offer a new way to think about the scaling issues associated with computation-hungry LLMs, according to computer scientist Jason Eshraghian at the University of California, Santa Cruz. "The brain has trillions of parameters, as do [newer] large language models," he says. "So if we understand how the brain works, then we should be able to scale up [using neuromorphic approaches] as well." In past work, he has developed a generative AI model based on SNNs. In April 2025, during the annual International Conference on Learning Representations, Eshraghian and his collaborators described an LLM adapted to run on a Loihi 2 chip from Intel—the first LLM, as far as they are aware, to be based on neuromorphic hardware. It was as accurate as a comparable graphics processing unit-based LLM but used half as much energy. Eshraghian doesn't think neuromorphic-based approaches will replace behemoths like those produced by Google or OpenAI; instead, he thinks they could work side by side, with behemoth LLMs offloading some tasks to neuromorphic models. He's also been analyzing hybrid models that may combine the best of all approaches.

Kudithipudi says it will be critical for researchers to find the best ways to leverage the strengths of neuromorphic approaches such as real-time, low-power monitoring, coupled with fast processing. In May 2025, for example, researchers at ETH Zürich reported on a real-time seizure monitor, built on neuromorphic hardware, that might help people with epilepsy (18). "If you're monitoring individuals in a smart hospital and using that data to make decisions on the fly, it could be helpful," says Kudithipudi. "Or in autonomous vehicles, where you have multiple sensors that are fusing information and predictions have to be made in less than a millisecond."

Gaining Acceptance, Slowly

A widespread embrace of neuromorphic computing won't happen overnight—and maybe not at all, cautions Zico Kolter, a computer scientist at Carnegie Mellon University in Pittsburgh, Pennsylvania. Although building a brain-like architecture onto a chip might enable it to function more efficiently, he doubts that today's market and chip-making industry would welcome the shift, given the huge investment required. There are other hurdles, too. Delbruck points out that neurons in the brain are plastic, which means they can change and adapt. To date, he says, "nobody has a neural model that has plastic synapses, in a practical way," though this does remain an active area of research.

Moreover, neuromorphic chips haven't yet found their killer app, one that no other AI system can do as well, and they can't just slide into today's AI systems or LLMs and instantly bring down their energy footprint, notes computational neuroscientist Brad Aimone at the Center for Computing Research at Sandia National Laboratories in Albuquerque, New Mexico, who develops algorithms for neuromorphic computers. Aimone and other researchers, though, expect there are ways to integrate neuromorphic chips through novel hardware designs and new, efficient algorithms. Already, recent smartphones have begun to incorporate neuromorphic-inspired ideas, such as sparsity, into their chip designs to increase speed and decrease energy usage. Delbruck believes that "many kinds of sparsity" will be exploited in future devices.

It's early days, but the advantages of the neuromorphic approach could portend a big impact in multiple arenas say some researchers. "I think neuromorphic computing is going to change the world," Aimone told a crowd in October 2024 during an Afternoon Lecture Series at NIH. "And it will do so in ways that most people don't appreciate yet."

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